ME551 Vibrations in Linear Systems

Spring Semester, 2025 **Department of Mechanical Engineering, KAIST**

Midterm Exam

1. (30) ***Fundamentals: True or False questions***

Answer to the **True or False** questions given below and explain briefly why. Without proper reasoning, no credit will be given even to the correct answers.

(a) (Linear System Analysis) Response to arbitrary excitation can be obtained by either convolution integral of impulse response function and input in the time domain; or multiplication of FRF with input in the frequency domain in both cases of free vibration and forced vibration. (5pts)

False. Convolution and FRF are required only for the force vibration. Initial conditions are required for free vibration.

(b) (State Equation) A second-order of linear MIMO systems with N-dof can be formulated into 1-st order N-dimensional state equation. (5pts)

False: It can be formulated into 1-st order 2N-dimensional state equation.

(c) (Principle of Virtual Work) In the Principle of Virtual Work, effects of internal and constraint forces are eliminated. (5pts)

True. Any reason is OK

(d) (Structural Damping) Structural damping yields the same energy loss per cycle with equivalent damping. It can be used in either transient or harmonic analysis (5pts)

False: It is valid only for harmonic excitation and corresponding harmonic response.

(e) (Eigenvalue Problem) For a real symmetric positive definite matrix A, we have real positive eigenvalues, and eigenvectors are also real-valued and orthogonal to each other. (5pts)

True. Any reason is OK

(f) (Modal Analysis) Gyroscopic asymmetric system cannot be diagonalized in modal analysis using right and left orthogonality of its eigenvectors. (5pt)

False: It can be diagonalized by right and left eigenvectors.

2. (10) *1st and 2nd Order Systems, Frequency Response Functions*

Motion of a third order system is described by a differential equation as follows:

$$\frac{d^3x}{dt^3}+3\frac{d^2x}{dt^2}+6\frac{dx}{dt}+8x(t)=f(t).$$

(a) Decompose the frequency response function of the above third order system ($G_3(j\omega)$) into a linear superposition of two sub-systems, *i.e.*, one of the first order ($G_1(j\omega)$) and the other of the second order ($G_2(j\omega)$). (5pts)

Solution

Apply Laplace transform with zero initial condition.

$$(s^3+3s^2+6s+8)X=F$$

$$G_3(s)$$

$$=\frac{1}{s^3+3s^2+6s+8}$$

$$=\frac{1}{(s+2)(s^2+s+4)}$$

$$=\frac{0.1667}{s+2}+\frac{-0.1667s+0.1667}{s^2+s+4}$$

$$\Rightarrow G_1(j\omega)=\frac{0.1667}{j\omega+2},\quad G_2(j\omega)=\frac{-0.1667(j\omega)+0.1667}{-\omega^2+(j\omega)+4}$$

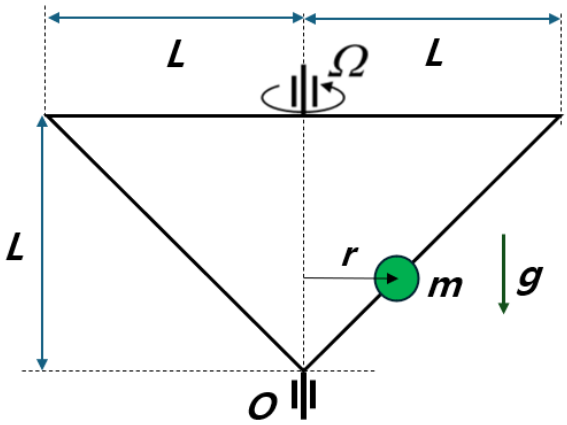
(b) A first order system is characterized typically by break frequency (ω_b) (or time constant) and an under-damped second order system is characterized typically by undamped natural frequency (ω_n) and damping ratio (ζ). Find out ω_b, ω_n and ζ. (5pts)

Solution

$$\omega_b=2\text{rad/s},\ \omega_n=2\,\text{rad/s},\ \zeta=0.25$$

3. **(25)** ***Lagrange Equation, Equilibrium Point, Linearization, Stability***

The system in Fig. P3 consists of a massless isosceles triangle of height L and width $2L$ rotating with a constant speed Ω about a vertical axis. A bead of mass m can slide freely along the triangle without friction. The position of the bead on the triangle is described by r and the gravity is given by g. The moving coordinate can be expressed by unit vectors $\hat{e}_r, \hat{e}_\phi, \hat{e}_k$ attached to the mass.



(a) Derive equation of motion of mass m with respect to r using the Lagrange equation. (10pts)

Solution

In cylindrical coordinates,

$$\mathbf{r} = r\,\hat{e}_r + r\,\hat{e}_k$$

$$\dot{\mathbf{r}} = \dot{r}\,\hat{e}_r + r\dot{\phi}\hat{e}_\phi + \dot{r}\,\hat{e}_k$$

$$= \dot{r}\,\hat{e}_r + r\Omega\hat{e}_\phi + \dot{r}\,\hat{e}_k$$

$$T = \frac{1}{2}m\left|\dot{\mathbf{r}}\right|^2 = m(\dot{r}^2 + \frac{1}{2}\Omega^2 r^2)$$

$$V = mgr$$

$$L = m(\dot{r}^2 + \frac{1}{2}\Omega^2 r^2) - mgr$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right) - \frac{\partial L}{\partial r} = 0$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}}\right) = 2m\ddot{r} \quad , \frac{\partial L}{\partial r} = m\Omega^2 r - mg$$

$$\therefore 2\ddot{r} - \Omega^2 r + g = 0$$

(b) Find out equilibrium position(s) of the mass denoted by r_0 and the required relationship between g , Ω which enables the mass to be in equilibrium for $0 < r_0 < L$. (5pts)

Solution

$$2\ddot{r} - \Omega^2 r + g = 0$$

$$\downarrow \ddot{r} = \dot{r} = 0 \;\; at \;\; r = r_0$$

$$-\Omega^2 r_0 + g = 0$$

$$r_0 = \frac{g}{\Omega^2}$$

$$0 < r_0 (= \frac{g}{\Omega^2}) < L \rightarrow \sqrt{\frac{g}{L}} < \Omega$$

(c) Linearize equation of motion for small motions around the equilibrium position r_0 obtained in (b). (5pts)

Solution

$$2\ddot{r} - \Omega^2 r + g = 0$$

$$\downarrow \begin{cases} r = r_0 + r_1(t) \\ r_0 = \dfrac{g}{\Omega^2} \end{cases}$$

$$2\ddot{r}_1 - \Omega^2 r_1 + g - \Omega^2 r_0 = 0$$

$$\Rightarrow 2\ddot{r}_1 - \Omega^2 r_1 + (g - \Omega^2 r_0) = 0$$

$$\Rightarrow \ddot{r}_1 - \frac{\Omega^2 r_1}{2} = 0$$

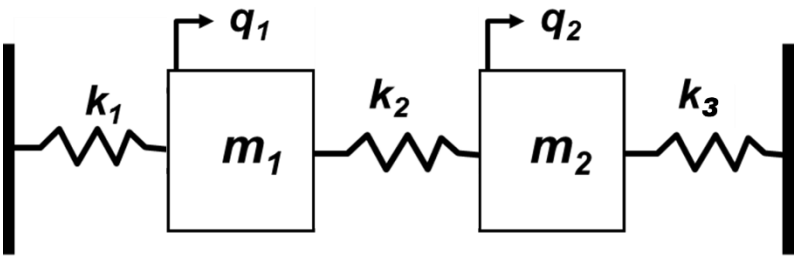
(d) Indicate the stability of the linearized system obtained in (c) by choosing, stable, asymptotically stable, and unstable and explain supporting reason for the stability characteristics. (5pts)

Solution

Unstable. (divergence, the stiffness $-\frac{\Omega^2}{2}$ has negative value)

4. **(35)** ***Multiple DOF Systems, Eigenvalue Problems, Modal Analysis***

Consider the free vibration of an undamped two-DOF system shown in Fig. P4, where the mass and the stiffness are given by $m_1 = m_2 = 9\,\text{kg}$, $k_1 = k_2 = k_3 = 36\,\text{N/m}$; and initial conditions $q_1(0) = q_2(0) = 1m,\ \dot{q}_1(0) = \dot{q}_2(0) = 0\,m/s$.



(a) Derive <u>matrix-vector equation of motion</u> of the free vibration $\boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{K}\boldsymbol{q} = \boldsymbol{0}$ with Lagrange Equation. Drive <u>the corresponding symmetric eigenvalue problem</u> assuming harmonic solution of natural vibration such that $\boldsymbol{q}(t) = \boldsymbol{u}e^{j\omega t}$. (5pts)

Solution

$$T = \frac{1}{2}m_1\dot{q}_1^{\,2} + \frac{1}{2}m_2\dot{q}_2^{\,2}$$

$$V = \frac{1}{2}k_1q_1^{\,2} + \frac{1}{2}k_2(q_2 - q_1)^2 + \frac{1}{2}k_3q_2^{\,2}$$

$$L = T - V = \frac{1}{2}m_1\dot{q}_1^{\,2} + \frac{1}{2}m_2\dot{q}_2^{\,2} - \frac{1}{2}k_1q_1^{\,2} - \frac{1}{2}k_2(q_2 - q_1)^2 - \frac{1}{2}k_3q_2^{\,2}$$

1) $\dfrac{d}{dt}\left(\dfrac{\partial L}{\partial \dot{q}_1}\right) - \dfrac{\partial L}{\partial q_1} = 0$

$\Rightarrow m_1\ddot{q}_1 + (k_1 + k_2)q_1 - k_2q_2 = 0$

2) $\dfrac{d}{dt}\left(\dfrac{\partial L}{\partial \dot{q}_2}\right) - \dfrac{\partial L}{\partial q_2} = 0$

$\Rightarrow m_2\ddot{q}_2 + (k_2 + k_3)q_2 - k_2q_1 = 0$

The above equation can be considered as $M\ddot{q} + Kq = Q$, then the natural frequencies and mode shape vectors can be obtained by solving eigenvalue problem, $\omega^2 Mu = Ku$.

$$\mathbf{M}=\begin{bmatrix}9 & 0\\ 0 & 9\end{bmatrix},\mathbf{K}=\begin{bmatrix}72 & -36\\ -36 & 72\end{bmatrix}$$

$$\therefore \begin{bmatrix}9 & 0\\ 0 & 9\end{bmatrix}\ddot{\mathbf{q}}+\begin{bmatrix}72 & -36\\ -36 & 72\end{bmatrix}\mathbf{q}=\mathbf{0}$$

$$[\mathbf{U},\mathbf{D}]=eig(\mathbf{K},\mathbf{M})$$

$$\mathbf{U}=\begin{bmatrix}-0.2357 & -0.2357\\ -0.2357 & 0.2357\end{bmatrix}$$

$$\uparrow\mathbf{u}_1 \qquad \uparrow\mathbf{u}_2$$

$$\mathbf{D}=\begin{bmatrix}4 & 0\\ 0 & 12\end{bmatrix}\rightarrow \omega_1=2,\ \ \omega_2=2\sqrt{3}$$

(b) Construct <u>standard eigenvalue problem</u> $\boldsymbol{Av}=\lambda\boldsymbol{v}$ ($\lambda=\omega^2$) from coordinate transform of $\boldsymbol{u}=\boldsymbol{M}^{-1/2}\boldsymbol{v}$ and obtain its two <u>eigenvalues</u> λ_1,λ_2, and <u>corresponding eigenvectors</u> $\boldsymbol{v}_1,\boldsymbol{v}_2$.(5pts)

Solution

$$\mathbf{A}=\mathbf{M}^{-1/2}\mathbf{K}\,\mathbf{M}^{-1/2}=\begin{bmatrix}8 & -4\\ -4 & 8\end{bmatrix}$$

$$\mathbf{V}=\begin{bmatrix}-0.7071 & -0.7071\\ -0.7071 & 0.7071\end{bmatrix}$$

$$\uparrow\mathbf{v}_1 \qquad \uparrow\mathbf{v}_2$$

$$\rightarrow \lambda_1=4,\lambda_2=12$$

(c) Calculate <u>modeshapes</u> $\boldsymbol{U} = [\boldsymbol{u}_1, \boldsymbol{u}_2]$ and <u>confirm the orthogonality</u> of modeshape vectors such that $\boldsymbol{U}^T \boldsymbol{M}\boldsymbol{U} = \boldsymbol{I}$ and $\boldsymbol{U}^T \boldsymbol{K}\boldsymbol{U} = diag(\lambda_1, \lambda_2)$. (5pts)

Solution

$$\mathbf{u}_1 = \mathbf{M}^{-1/2}\mathbf{v}_1 = \begin{bmatrix} -0.2357 \\ -0.2357 \end{bmatrix}$$

$$\mathbf{u}_2 = \mathbf{M}^{-1/2}\mathbf{v}_2 = \begin{bmatrix} -0.2357 \\ 0.2357 \end{bmatrix}$$

$$\mathbf{U} = \begin{bmatrix} -0.2357 & -0.2357 \\ -0.2357 & 0.2357 \end{bmatrix}$$

$$\mathbf{U}^T\mathbf{M}\,\mathbf{U} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\mathbf{U}^T\mathbf{K}\,\mathbf{U} = \begin{bmatrix} 4 & 0 \\ 0 & 12 \end{bmatrix}$$

(d) Add the proportional damping $\boldsymbol{C} = 0.2\boldsymbol{M} + 0.1\boldsymbol{K}$ into the system such that the matrix-vector equation of motion of the free vibration becomes $\boldsymbol{M}\ddot{\boldsymbol{q}} + \boldsymbol{C}\dot{\boldsymbol{q}} + \boldsymbol{K}\boldsymbol{q} = \boldsymbol{0}$. Construct <u>decoupled damped free vibration equation</u> in terms of modal coordinates $\eta_1(t), \eta_2(t)$ using the modal expansion as $\boldsymbol{q}(t) = \boldsymbol{u}_1\eta_1(t) + \boldsymbol{u}_2\eta_2(t)$. (5pts)

Solution

$$\mathbf{M}\ddot{\mathbf{q}} + \mathbf{C}\dot{\mathbf{q}} + \mathbf{K}\mathbf{q} = \mathbf{0}$$

$$\downarrow \mathbf{q} = \mathbf{U}\boldsymbol{\eta}$$

$$\mathbf{U}^{\mathbf{T}}\mathbf{M}\mathbf{U}\ddot{\boldsymbol{\eta}} + 0.2\mathbf{U}^{\mathbf{T}}\mathbf{M}\mathbf{U}\dot{\boldsymbol{\eta}} + 0.1\mathbf{U}^{\mathbf{T}}\mathbf{K}\mathbf{U}\dot{\boldsymbol{\eta}} + \mathbf{U}^{\mathbf{T}}\mathbf{K}\mathbf{U}\boldsymbol{\eta} = 0$$

$$\Rightarrow \begin{bmatrix} \ddot{\eta}_1 \\ \ddot{\eta}_2 \end{bmatrix} + \begin{bmatrix} 0.2 + 0.1\omega_1^2 & \\ & 0.2 + 0.1\omega_2^2 \end{bmatrix} \begin{bmatrix} \dot{\eta}_1 \\ \dot{\eta}_2 \end{bmatrix} + \begin{bmatrix} \omega_1^2 & \\ & \omega_2^2 \end{bmatrix} \begin{bmatrix} \eta_1 \\ \eta_2 \end{bmatrix} = 0$$

(e) Obtain the modal response $\eta_1(t), \eta_2(t)$ using the SDOF free vibration solution with an initial displacement η_0 and initial velocity $\dot{\eta}_0$. (10pts)

Solution

$$q_1(0) = q_2(0) = 1$$
$$\dot{q}_1(0) = \dot{q}_2(0) = 0$$

$$\mathbf{U}^T\mathbf{M}\mathbf{q}_0 = \mathbf{U}^T\mathbf{M}\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}-4.2426\\0\end{bmatrix} = \boldsymbol{\eta}_0$$

$$\mathbf{U}^T\mathbf{M}\dot{\mathbf{q}}_0 = \begin{bmatrix}0\\0\end{bmatrix} = \dot{\boldsymbol{\eta}}_0$$

$$\ddot{\boldsymbol{\eta}} + \begin{bmatrix}0.2+0.1\omega_1^2 & \\ & 0.2+0.1\omega_2^2\end{bmatrix}\dot{\boldsymbol{\eta}} + \begin{bmatrix}\omega_1^2 & \\ & \omega_2^2\end{bmatrix}\boldsymbol{\eta} = 0$$

$$\therefore \eta_1(t) = e^{-0.3t}(-4.2426\cos(1.9763t) - 0.644\sin(1.9763t)),$$
$$\eta_2(t) = 0$$

(f) Obtain the physical solution $\boldsymbol{q}(t) = \boldsymbol{u}_1\eta_1(t) + \boldsymbol{u}_2\eta_2(t)$ from the modal solution $\eta_1(t), \eta_2(t)$ obtained in (d). (5pts)

Solution

$$\begin{bmatrix}q_1(t)\\q_2(t)\end{bmatrix} = \begin{bmatrix}u_1 & u_2\end{bmatrix}\begin{bmatrix}\eta_1(t)\\\eta_2(t)\end{bmatrix}$$
$$= \begin{bmatrix}-0.2357 & -0.2357\\-0.2357 & 0.2357\end{bmatrix}\begin{bmatrix}e^{-0.3t}(-4.2426\cos(1.9763t) - 0.644\sin(1.9763t))\\0\end{bmatrix}$$
$$= \begin{bmatrix}e^{-0.3t}(\cos(1.9763t) + 0.1518\sin(1.9763t))\\e^{-0.3t}(\cos(1.9763t) + 0.1518\sin(1.9763t))\end{bmatrix}$$

q_1 and q_2 have only the first mode.

Bonus Problem

(5pts) Please make a mechanical vibration problem by yourself and show the solution procedure. Any problem ideated from ME551 and proper answer will be fully credited.